Technical Requirement Document (TRD)

CampusFind Pro - System Architecture, Database Schemas, and Environment Configs

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1. System Architecture Overview

CampusFind Pro is developed on a lightweight, modular MVC-based architectural model. The backend runs on PHP 8+ using native cURL libraries and driver extensions, communicating securely with MongoDB Atlas NoSQL database clusters. The frontend is styled using Bootstrap 5 and custom vanilla CSS variables, rendering layouts dynamically. Outgoing communication uses HTTPS port 443 to make REST requests to Resend and Brevo APIs.

2. MongoDB Atlas Schema Definitions

The NoSQL database stores all document models dynamically under the following collections:

- **users:** stores email (string), name (string), avatar (string url or path), role (student/admin), and verification_code (null/string).
- **lost_items / found_items:** stores title (string), description (text), category_id (BSON ObjectId), user_id (BSON ObjectId), location (string), lost_date/found_date (string), reward (float), image (string filename), and status (lost/claimed).
- **claims:** stores claimer_id (BSON ObjectId), item_id (BSON ObjectId), item_type (string), status (pending/approved/rejected), admin_notes (text), and created_at (date).
- **categories:** stores preseeded labels like Electronics, Documents, Keys, and Books.
- **support_messages:** stores user inquiries submitted via contact forms (name, email, message, status, and created_at).

3. Recursive BSON Query Parser (castFilterIds)

To resolve type incompatibilities between standard PHP sessions (which store user ID strings) and MongoDB NoSQL storage (which uses BSON ObjectIds), we implemented a recursive parser inside the Database class. The helper method (castFilterIds) automatically scans all input query arrays. If it finds a 24-character hexadecimal string mapped to keys ending in _id, claimer_id, or item_id, it converts it to a BSON ObjectId on the fly, avoiding manual database query crashes across the app.

4. Server Environment Requirements

The production environment requires PHP 8+, PHP-cURL enabled, the mongodb php extension, and the loading of environment keys (MONGODB_URI, MONGODB_DB, GOOGLE_CLIENT_ID, GOOGLE_CLIENT_SECRET, BREVO_API_KEY, BREVO_SENDER_EMAIL) via Apache server configurations or Render dashboard settings.